

MARIA LISA D SILVA

Data Engineer

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SUMMARY

Databricks Certified Data Engineer Professional (SnowPro Core, AWS Cloud Practitioner certifications) with **1+** year of enterprise experience building cloud-native Lakehouse platforms using **Databricks, Apache Spark, Airflow, Talend, SQL and Python**. Skilled in **Medallion architecture, large-scale ETL/ELT pipelines, data migrations, API ingestion, dbt, DuckDB and GitHub Actions CI/CD**. Hands-on in DAG debugging, data validation and root cause analysis, delivering 98%+ data accuracy in production-grade systems.

TECHNICAL SKILLS

Programming Languages: SQL, Python, Bash, PySpark, Spark SQL, Java, C/C++

Data Engineering: Databricks, Apache Spark, Delta Lake, Lakehouse and Medallion Architecture, Data Modeling, Data Migration

Pipelines & Orchestration: Apache Airflow, Talend, dbt, Dagster, ETL/ELT, API Ingestion, Incremental Loading

Databases & Data Warehousing: PostgreSQL, Greenplum, Oracle, MySQL, Snowflake, DuckDB, MotherDuck

Cloud, DevOps & Version Control: AWS (S3), Git, GitHub, GitHub Actions, CI/CD, Cloudflare Pages

Data Processing & BI: Pandas, NumPy, Scikit-learn, Parquet, Power BI, Tableau, DAX, Excel

Other Tools: pytest, mypy, DBeaver, Postman, VS Code, Jupyter, Google Colab, Spring Boot, REST APIs

PROFESSIONAL EXPERIENCE

Data Engineer | Genpact, Bengaluru, India

Oct 2024 – Dec 2025

- Engineered **Spark SQL ELT migrations** of **900+ Greenplum/PostgreSQL objects** across **15+ domains** to **Databricks Lakehouse (Medallion Architecture)**, owning **300+ end to end**: DDL conversion, schema mapping, transformations, views, data validation.
- Migrated **PXF/reverse-PXF** external tables and data exchange workflows, recreating Greenplum read/write patterns using **Spark SQL**, Databricks **external tables** and **AWS S3** storage across Oracle, MySQL, Smartsheet, Box and other enterprise sources.
- Modernized legacy **Talend TAC ingestion/egress** workflows into scheduled **Airflow DAGs** with **YAML-based GitHub CI/CD** and **DBeaver metadata** configurations, boosting pipeline automation by **~80%**.
- Achieved **98%+ data accuracy** via Spark pipeline validation, **SQL reconciliation**, log diagnostics, **root cause analysis (RCA)** and **SIT testing**, resolving schema, transformation and data issues with business stakeholders.
- Awarded **Bronze Performance Excellence** and selected as a **Top 20 Performer** for Technical Leadership Development.

GenAI Intern | Genpact, Bengaluru, India

Feb 2024 – Jul 2024

- Built an **AI-powered document deduplication system** (NLTK, TF-IDF, LLMs, Sentence Transformers, Affinity Propagation), reducing average document length by **54.6%** and eliminating .docx duplicates (**0.97 semantic similarity**).
- Developed a **Vehicle Financing Performance Dashboard** with Python ML models (**85% accuracy**), SQL/DAX data cleaning, feature engineering, visualization and KPIs, raising loan approvals by **20%** and cutting defaults by **25%**.

Software Development Intern | NMBR Systems Private Limited, Bengaluru, India

Jun 2023 – Jul 2023

- Built a **Java/Spring Boot management system** with MySQL, REST APIs and Postman-tested CRUD operations.
- Optimized SQL queries and implemented JPA/Hibernate ORM with Java Streams, reducing manual effort.

Web Development Intern | D'Souza Metal Cutting Private Limited, Mumbai, India

Jun 2022 – Aug 2022

- Developed a responsive HTML/CSS UI for PDF invoice generation, cutting manual processing time by **50%**.

PROJECTS

HarbourOS - AIS Port-Call Data Pipeline | *Python, SQL, DuckDB, MotherDuck, dbt, GitHub Actions, Dagster, pytest, Parquet*

- Built **Bronze/Silver/Gold ELT pipeline** ingesting **500K+** AIS messages/day from **4,000+** ships via micro-batches, watermark-based incremental loads.
- Generated analytics-ready port-call events by mapping vessels to 3,761 berths with rule-based confidence scoring, **dbt star schema** with **data-quality tests**, row-level quarantine, and **zero-loss reconciliation**.
- Automated CI/CD with self-chaining GitHub Actions, schema-drift recovery, failure alerts, and **99.8%** fewer database round trips through Parquet batching, developed using AI-assisted workflows with Claude Code.

Crime in India Data Platform (2001-2024) | *Python, SQL, DuckDB, dbt, GitHub Actions, pytest, scikit-learn*

- Built batch ELT pipeline processing **86 checksum-verified** NCRB files into analytics-ready datasets covering **63.6M** crime records.
- Designed **dbt star schema** with **87 tests, 75 quality checks**, ensuring reliability via reconciliation and automated CI/CD validation.

CERTIFICATIONS

Databricks Certified Data Engineer Professional | Databricks Certified Data Engineer Associate | **SnowPro Core** | **AWS Certified Cloud Practitioner** | Databricks Generative AI Fundamentals | Databricks AI Agent Fundamentals

EDUCATION

B.Tech, Computer & Communication Engineering (Minor: Big Data Analytics) | Manipal Institute of Technology

2020 – 2024